Theorem 5.2 (Correctness of Source-DPOR). *For all execution sequences E, Algorithm 1 explores some execution sequence E' which is in $[E.v]_{\simeq}$ for some v. In particular, for all maximal execution sequences E, Algorithm 1 explores some execution sequence E' which is in $[E]_{\simeq}$.*

The proof is given at the end of this section, after establishing a series of intermediate lemmas. We begin by introducing some notation. Let:

- *init_sleep*(E) be the value of *Sleep* when calling *Explore*$(E, Sleep)$,
- *final_sleep*(E) be the value of *Sleep* when *Explore*$(E, Sleep)$ returns,
- *done*(E) be *final_sleep*$(E) \setminus$ *init_sleep*(E), i.e., the set of processes that are explored from $s_{[E]}$ (added in *Sleep* by line 12),
- *to_be_explored*(E) be the set $\{w \mid E \vdash w \wedge (I_{[E]}(w) \cap init_sleep(E) = \emptyset)\}$ of possible continuations w of E such that $I_{[E]}(w) \cap init_sleep(E) = \emptyset$; intuitively, this is the set of sequences that should be explored by the call *Explore*$(E, Sleep)$,
- $e \precsim_E p$ denote that p is not blocked after E and that $e \precsim_{E.p} next_{[E]}(p)$.

We are now ready to state the lemma that will eventually be used to prove correctness.

Lemma 5.3. *Whenever a call to Explore$(E, Sleep)$ returns during Algorithm 1, then for all sequences $w \in$ to_be_explored(E), the algorithm has explored some execution sequence $E.w'$ which is in $[E.w.v]_{\simeq}$ for some v.*

Proof. The proof is by induction on the order in which states (i.e., execution sequences) are backtracked by the algorithm. The first time *Explore* returns, it has just explored a maximal sequence, so the base case holds trivially. Consider a sequence E. As inductive hypothesis, we assume that the statement in the lemma holds for all execution sequences $E.p$ with $p \in done(E)$. Whenever we invoke the induction hypothesis, we need to establish a condition of form $w' \in to_be_explored(E.p)$ for some process p and sequence w'. We will refer to this as the IH-condition, and try to separate its establishment from the rest of the flow in the proof. We structure the proof of the lemma as a sequence of *claims*.

Claim 5.4. *done(E) is a source set for to_be_explored(E) after E.*

By the definition of source sets (Definition 4.3), this claim states that for each $w \in to_be_explored(E)$, there is some $p \in done(E)$ such that $p \in WI_{[E]}(w)$. Since, by the definition of *to_be_explored*(E), the set of sequences w with $E \vdash w$ is the union of *to_be_explored*(E) and the set $\{w \mid E \vdash w \wedge init_sleep(E) \neq \emptyset\}$, it implies that the set *final_sleep*(E) is a source set for $\{w \mid E \vdash w\}$.

Proof of Claim 5.4. By contradiction. Assume a $w \in to_be_explored(E)$, such that for all $p \in done(E)$ we have $p \notin WI_{[E]}(w)$. We will prove that such a sequence cannot exist.

If *done*(E) is empty, no process was explored from that state: this can only happen if the check on line 2 has failed (otherwise process p on that line will later be added by line 12) which contradicts that *to_be_explored*(E) is non-empty. Therefore, *done*(E) is not empty.

For each process $p \in done(E)$, let w_p be the longest prefix of w such that $E \vdash p \diamond w_p$, and let e_p be the first event in $dom_{[E]}(w)$ which is not in w_p. We have:

$$p \neq \widehat{e_p} \qquad \text{(otherwise from } E \vdash p \diamond w_p \text{ we derive } p \in WI_{[E]}(w)) \tag{7}$$

Let q be such that w_q is a longest prefix among the prefixes w_p for $p \in done(E)$. If there are several processes p such that w_p is the same longest prefix, then let q be the process among these which is explored first from $s_{[E]}$. Finally, let w_R be $w_q.\widehat{e_q}$ (Fig. 6).